

Figure 1: **Late Crystallization**: Correct answers remain invisible in logit space throughout intermediate layers, then abruptly “crystallize” into top predictions at the final layer.

MECHLENS: Late Crystallization of Factual Knowledge Explains Intervention Effectiveness in Language Models

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Abstract

Understanding where LLMs store factual knowledge is critical for hallucination mitigation. We discover **Late Crystallization**: factual knowledge does not gradually emerge across layers but “crystallizes” abruptly at the final layers. In Qwen2.5-7B, 85.9% of correct answers *never* enter top-10 predictions at any intermediate layer. Cross-architecture validation (Llama-3.1-8B: 71.0%, Mistral-7B: 27.1%) confirms late emergence (>80% depth) is universal, while strict final-layer crystallization varies with attention mechanism. Cross-scale (Qwen2.5-14B: 95.8% FEP Depth) and cross-benchmark (MMLU: 98.2%) validation confirm generality; tuned lens comparison rules out probe artifacts ($\Delta=0.2\%$).

Late Crystallization explains an **architecture-dependent intervention pattern**: CAA outperforms DOLA on moderate-crystallization models (Llama: +33.5% vs. +1.9%), while DOLA dominates on high-crystallization Qwen (+25.4% MC1). LayerNorm ablation shows crystallization is intrinsic to the residual stream; LN scaling ($\times 1.2$) yields +11.8% MC1 with zero overhead. We also reveal a **Computability–Memorization Spectrum**: computable knowledge crystallizes earlier (layer 22.1/28) than memorized facts (28.0/28). We release MECHLENS supporting five model families.¹

1 Introduction

Large language models (LLMs) frequently produce plausible-sounding but factually incorrect outputs—hallucinations—posing significant challenges for deployment in high-stakes applications (Huang et al., 2023; Ji et al., 2023). A natural approach to mitigating hallucinations is to identify the internal components responsible for factual recall and intervene directly on their activations. However, the effectiveness of such interventions depends critically on whether factual knowledge is *localized* in specific model components or *distributed* across the network.

Prior work on knowledge localization has produced mixed evidence. Causal tracing (Meng et al., 2022) identifies middle-layer MLPs as critical for factual recall in GPT-2, motivating localized editing methods like ROME and MEMIT. However, recent work suggests knowledge may be more distributed than initially assumed (Geva et al., 2022), raising questions about whether activation-level interventions can effectively modulate factual behavior.

¹Code: <https://anonymous.4open.science/r/MechLens-COLM2026>

	Prior Work	This Work
Observation	Qualitative	Quantified (85.9%)
Metric	None	FEP (formal)
Cross-arch	Limited	3 architectures
Causal test	None	LN ablation
Intervention link	None	Predictive theory

Table 1: Comparison with prior logit lens studies.

We introduce MECHLENS, a unified mechanistic interpretability framework that enables systematic investigation of these questions. Through comprehensive experiments combining logit lens analysis, layer contrasting, and activation interventions, we discover a phenomenon we term **Late Crystallization**: factual knowledge does not gradually emerge in intermediate layers but crystallizes abruptly at the final layers (Figure 1). Our contributions:

- **Late Crystallization**: 85.9% of correct answers never enter top-10 at any intermediate layer, validated across architectures, scales (7–14B), benchmarks (TruthfulQA, MMLU), and probing methods (logit lens vs. tuned lens, $\Delta=0.2\%$)
- **Architecture-dependent intervention pattern**: CAA outperforms DOLA on moderate-crystallization models ($p<0.001$); DOLA dominates on high-crystallization Qwen (+25.4% MC1)
- **Causal evidence**: LayerNorm ablation shows crystallization is intrinsic to the residual stream; LN scaling ($\times 1.2$) yields +11.8% MC1 with zero overhead
- **Computability–Memorization Spectrum**: Computable knowledge crystallizes at layer 22.1/28; memorized facts only at 28.0/28

Together, these results bridge language model *understanding* (when does factual knowledge become explicit?), *theory* (why do different interventions succeed or fail?), and *practice* (how should practitioners select interventions based on architecture?).

2 Related Work

Knowledge Localization and Editing. Causal tracing (Meng et al., 2022) identifies middle-layer MLPs as critical for factual recall in GPT-2, motivating ROME and MEMIT (Meng et al., 2023). Geva et al. (2021; 2022) show MLPs function as key-value memories. Circuit tracing (Anthropic, 2025) reveals distributed factual circuits, while logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023) enable layer-wise prediction tracking. Concurrent work (Wang, 2025) extends logit lens tooling to modern architectures and observes late-layer knowledge concentration. Table 1 summarizes how our work extends beyond prior observations.

Activation Intervention for Factual Accuracy. ITI (Li et al., 2023) shifts activations along probed truthfulness directions. DOLA (Chuang et al., 2024) contrasts early/late logit distributions without learned directions. CAA (Turner et al., 2023) and representation engineering (Zou et al., 2023) steer via contrastive activation vectors. SADI (Wang et al., 2025) achieves state-of-the-art MC1=67% via semantic-adaptive intervention on instruction-tuned models. CCS (Burns et al., 2023) discovers latent knowledge through consistency constraints. These methods operate in activation space (ITI, CAA, SADI) or logit space (DOLA); our work provides a mechanistic account of *why* their effectiveness differs across architectures.

3 The MECHLENS Framework

MECHLENS is built on TransformerLens (Nanda & Bloom, 2022) and supports five model families with diverse architectures (Table 2).

Model	Layers	Heads	d	Attention
Qwen2.5-0.5B	24	14	896	GQA (2 KV)
Qwen2.5-7B	28	28	3584	GQA (4 KV)
Qwen2.5-14B	48	40	5120	GQA (8 KV)
Pythia-1.4B	24	16	2048	MHA
Llama-3.1-8B	32	32	4096	GQA (8 KV)
Mistral-7B	32	32	4096	GQA+SWA (8 KV)

Table 2: Supported model architectures. Qwen2.5-14B enables cross-scale validation (§7.2).

Analysis. We extend causal tracing (Meng et al., 2022) with adaptive noise calibration ($\sigma = \alpha \cdot \text{std}(\mathbf{E}(x))$, $\alpha=10$), KL divergence metrics, and multi-run averaging ($n=5$). We also implement contrastive activation analysis, computing normalized L2 distance between correct and hallucinated residual stream activations at each layer. Full methodological details are in Appendix B.

Intervention. MECHLENS supports three reversible intervention types through hooks: ablation ($\mathbf{h}'_{l,c} = \mathbf{0}$), scaling ($\mathbf{h}'_{l,c} = \alpha \cdot \mathbf{h}_{l,c}$), and injection ($\mathbf{h}'_{l,c} = \mathbf{h}_{l,c}^{\text{source}}$). For CAA, we follow Turner et al. (2023): given paired truthful/untruthful prompts, we extract contrastive directions $\mathbf{d}_l = \frac{1}{N} \sum_i (\mathbf{h}_l(x_i^+) - \mathbf{h}_l(x_i^-))$ and add $\alpha \cdot \mathbf{d}_l$ to the top- k layers ranked by $\|\mathbf{d}_l\|_2$ at inference time. Directions are extracted from 256 TruthfulQA training pairs.

4 Experimental Setup

Models and Phases. *Phase 1* (causal tracing, contrastive analysis, activation scaling) uses Qwen2.5-0.5B and Pythia-1.4B—matched 24-layer models with different architectures (GQA/SwiGLU vs. MHA/GELU). *Phase 2* (FEP detection, MC1/MC2 evaluation, cross-architecture validation) uses Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B on the full TruthfulQA benchmark (817 samples) (Lin et al., 2022). *Phase 3* (cross-scale validation) extends FEP detection to Qwen2.5-14B (48 layers) to test whether Late Crystallization persists at larger scale.

Evaluation. Phase 1 screening uses 200 TruthfulQA samples with keyword matching plus LLM-as-Judge (Zheng et al., 2023). Phase 2 uses standardized MC1 (single-correct accuracy) and MC2 (normalized probability mass) on all 817 samples. We test 20 activation scaling strategies across 6 categories plus an 18-configuration ITI grid search; full details and results are in Appendix A.

5 Results

5.1 Negative Results: Activation Scaling and ITI

Simple activation scaling universally fails across 20 strategies and 4 models ($\leq 2\%$ gain; up to 28% degradation), and an 18-configuration ITI grid search achieves at most +2%. This consistency across 38 configurations demands a mechanistic explanation (§6; full tables in Appendix A).

5.2 MC1/MC2 Evaluation with Advanced Interventions

We evaluate four methods on the full 817-sample TruthfulQA benchmark using Qwen2.5-7B (Table 3).

Key findings. A clear hierarchy emerges: DoLA dynamic (+25.4% MC1) > CAA top_k=10 (+15.5%) > ITI top_k=10 (+10.0%) > simple scaling (0%). Both ITI and CAA show zero improvement at top_k $\in \{3, 5\}$ but substantial gains at top_k=10, indicating a layer-count

Method	Configuration	MC1	MC2
Baseline	—	0.2215	0.3921
ITI	top_k=3, $\lambda=1-3$	0.2215	0.3921
ITI	top_k=5, $\lambda=1-3$	0.2203	0.3926
ITI	top_k=10, $\lambda=3$	0.2436	0.4178
DoLA	early (static)	0.2326	0.4371
DoLA	mid (static)	0.2448	0.4555
DoLA	dynamic	0.2778	0.4822
CAA	top_k=3, all coeff	0.2215	0.3921
CAA	top_k=5, all coeff	0.2203	0.3922
CAA	top_k=10, coeff=5.0	0.2558	0.4338

Table 3: MC1/MC2 on TruthfulQA (Qwen2.5-7B, 817 samples). DoLA dynamic achieves the best MC1 (+25.4% over baseline, $p=0.009$). 95% bootstrap CI for MC1 $\approx \pm 0.03$ ($n=817$).

Probe	Mean FEP	Late Crystal	Exact Match	± 2 Match
Logit lens	27.3 $\pm$ 1.8	85.9%	74.9%	76.4%
Tuned lens	25.1 $\pm$ 7.4	85.7%		

Table 4: Tuned lens vs. logit lens FEP (Qwen2.5-7B, 817 samples). Both yield near-identical late crystallization rates ($\Delta=0.2\%$).

threshold. That DoLA (logit-space) achieves the largest gains motivates our mechanistic analysis in Section 6. All experiments use base (non-instruction-tuned) models (MC1 baselines: 18.9–22.2%); absolute values should not be compared with instruction-tuned methods like SADI (Wang et al., 2025) (MC1=67%). DoLA’s improvement is significant (two-proportions z-test: $z=2.63$, $p=0.009$; survives Bonferroni correction at $\alpha=0.017$).

6 Late Crystallization of Factual Knowledge

The effectiveness hierarchy observed in Section 5.2—and the striking failure of simple activation scaling—demands a mechanistic explanation. We hypothesize and test the idea that factual knowledge in LLMs undergoes a phase transition at the final layers, which we term **Late Crystallization**.

6.1 Factual Emergence Point Detection

We define the **Factual Emergence Point (FEP)** for a query q with correct answer a as the earliest layer L at which a enters the top- k predictions under logit lens projection:

$$L_{\text{FEP}} = \min\{L : \text{rank}(a, \mathbf{W}_U \cdot \text{LN}(\mathbf{h}_L)) \leq k\} \quad (1)$$

where $\mathbf{h}_L$ is the residual stream at layer L , LN is the final layer norm, and $\mathbf{W}_U$ is the unembedding matrix. If a never enters top- k at any intermediate layer, we set $L_{\text{FEP}} = L_{\text{max}}$ (the final layer). We use $k = 10$ throughout.

Tuned lens validation. A natural concern is that $\text{FEP} = L_{\text{max}}$ for 85.9% of samples could reflect logit lens’s known limitations rather than a genuine phenomenon. We train per-layer affine probes on 2,000 WikiText-2 samples and compute tuned lens FEP on all 817 TruthfulQA samples: the tuned lens yields 85.7% late crystallization—within 0.2 percentage points of the logit lens (85.9%), with 74.9% of samples having identical FEP (Table 4). Three further lines of evidence argue against a probe artifact: (1) LayerNorm ablation produces identical FEP distributions (§7.5); (2) a systematic cross-architecture gradient (Qwen 85.9% > Llama 71.0% > Mistral 27.1%) correlates with attention mechanisms rather than showing random variation; (3) crystallization degree *predicts* which intervention dominates (§7).

FEP Layer	Count	Percentage
28 (final / never)	702	85.9%
23	47	5.8%
21	20	2.4%
25	13	1.6%
Other ($\leq 1.2\%$ each)	35	4.3%

Table 5: FEP distribution (Qwen2.5-7B, 817 TruthfulQA samples). 85.9% of correct answers *never* enter top-10 at any intermediate layer.

130 We compute FEP for all 817 TruthfulQA samples on Qwen2.5-7B (28 layers), tracking the
131 rank of each correct answer’s first token across all layers via logit lens projection.

132 **Robustness to k threshold.** Late Crystallization is robust across $k \in \{1, 3, 5, 10, 20, 50, 100\}$:
133 even at $k=100$ (top 0.1% of vocabulary), 64.7% of correct answers still never appear at any
134 intermediate layer, and FEP Depth remains above 93% (Table 14 in Appendix F).

135 6.2 The Late Crystallization Phenomenon

136 Table 5 reveals a striking pattern: **85.9% of correct answers (702/817) never enter the top-**
137 **10 predictions at any intermediate layer** (mean FEP = 27.3 ± 1.8 out of 28 layers). This
138 **Late Crystallization** shows that factual knowledge does not gradually “build up” across
139 depth but exists in a distributed, implicit form within the residual stream before abruptly
140 crystallizing into explicit predictions at the final layer.

141 6.3 Computability–Memorization Spectrum

142 FEP varies systematically across knowledge categories (full table in Appendix H): Logi-
143 cal Falsehood crystallizes earliest (mean FEP=22.1, $\sigma=2.6$), while categories like History,
144 Psychology, and Weather show FEP=28.0 with *zero* variance—every sample has its answer
145 invisible until the final layer. This **Computability–Memorization Spectrum** reveals that
146 “computable” knowledge emerges at intermediate layers, while purely memorized world
147 knowledge remains fully distributed until crystallization.

148 6.4 FEP–DoLA Decorrelation

149 DoLA’s dynamic layer selection does not correlate with FEP (Pearson $r = -0.057$, $p = 0.103$):
150 DoLA selects near-exclusively layer 0–1 (mean = 0.95), while FEP concentrates at layer 27–28.
151 This reveals DoLA’s true mechanism: contrasting the maximally early layer against the
152 final layer, amplifying the factual signal that crystallization produces. Late Crystallization
153 explains all observed patterns: DoLA contrasts pre- vs. post-crystallization logits; CAA/ITI
154 require $\text{top}_k \geq 10$ to span the pre-crystallization window; simple scaling amplifies all
155 components uniformly, unable to selectively steer the nonlinear crystallization process.

156 7 Cross-Architecture Validation and Causal Analysis

157 We extend our analysis to Llama-3.1-8B (32 layers, GQA) and Mistral-7B (32 layers,
158 GQA+SWA).

159 7.1 Cross-Architecture FEP Detection

160 Table 6 and Figure 2 reveal two key findings: (1) **Late knowledge emergence is universal**—
161 across all four models (including 14B), factual knowledge emerges at $>80\%$ model depth;
162 (2) **The degree varies with architecture**—strict final-layer crystallization ranges from 85.9%
163 (Qwen-7B) to 27.1% (Mistral), where sliding window attention appears to enable earlier
164 information routing.

Model	Layers	Mean FEP	FEP Depth	Late Crystal
Qwen2.5-7B	28	27.3 ± 1.8	97.5%	85.9%
Qwen2.5-14B	48	46.0 ± 4.9	95.8%	77.7%
Llama-3.1-8B	32	29.4 ± 4.9	91.9%	71.0%
Mistral-7B	32	26.3 ± 6.2	82.3%	27.1%

Table 6: Cross-architecture and cross-scale FEP detection (817 samples). FEP Depth = Mean FEP / total layers. All models show knowledge emergence >80% depth.

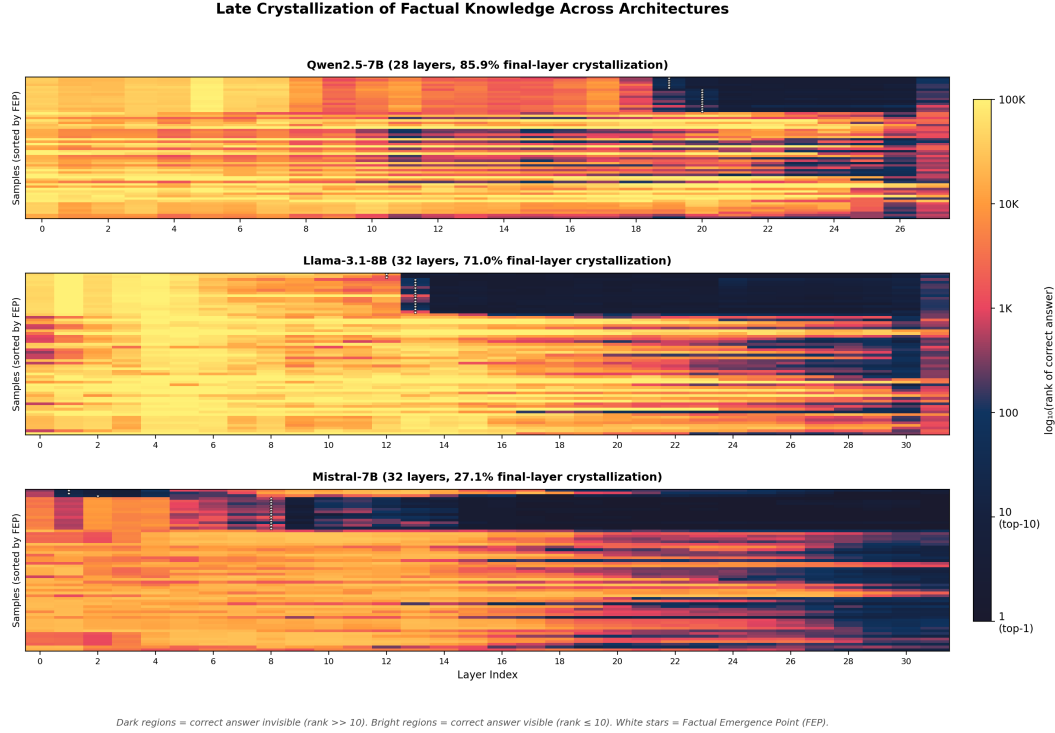


Figure 2: Late Crystallization across architectures. Each row: a TruthfulQA sample; each column: a layer. Color: $\log_{10}(\text{rank})$ of correct answer under logit lens. White stars: FEP.

7.2 Cross-Scale Validation (Qwen2.5-14B)

To test scale invariance, we run FEP detection on Qwen2.5-14B (48 layers, 14.7B parameters). Results (Table 6, row 2) confirm Late Crystallization *persists*: 77.7% of correct answers never enter top-10 at any intermediate layer, and FEP Depth is 95.8% (vs. 97.5% at 7B). The negligible 1.7pp difference in FEP Depth confirms that the relative crystallization layer is scale-invariant within the same architecture family.

7.3 Cross-Benchmark Validation (MMLU)

To verify generality beyond TruthfulQA, we run FEP detection on 1,200 MMLU samples spanning 24 subjects. Late Crystallization is even *more* pronounced: 98.2% of correct answers never enter top-10 at any intermediate layer, with near-perfect consistency across STEM (99.1%), Humanities (98.0%), Social Sciences (97.2%), and Other (98.0%) domains (Table 15 in Appendix G). The higher MMLU crystallization is consistent with MMLU testing primarily memorized domain knowledge, while TruthfulQA includes “computable” categories that emerge earlier.

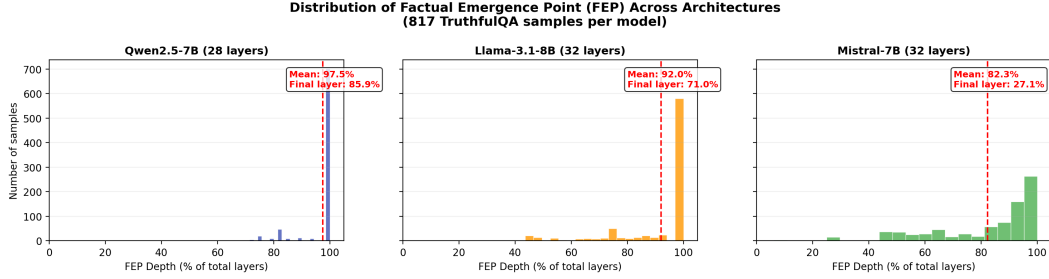


Figure 3: FEP depth distribution across architectures (817 samples each). Qwen concentrates at the final layer (85.9%), Llama shows a bimodal pattern (71.0%), Mistral distributes broadly (27.1%) while maintaining >82% mean depth.

Model	Method	MC1	$\Delta\%$
Llama-3.1-8B	Baseline	0.1897	—
	DoLA dynamic	0.1934	+1.9
	CAA (top_k=10)	0.2534	+33.5
Mistral-7B	Baseline	0.2044	—
	DoLA dynamic	0.2277	+11.4
	CAA (top_k=10)	0.2705	+32.3
Qwen2.5-7B	Baseline	0.2215	—
	DoLA dynamic	0.2778	+25.4
	CAA (top_k=10)	0.2558	+15.5

Table 7: Cross-architecture intervention results (817 samples). CAA outperforms DoLA on moderate-crystallization models; DoLA achieves largest gains on high-crystallization Qwen.

7.4 Cross-Architecture Intervention Hierarchy

Table 7 reveals an *architecture-dependent* pattern: on moderate-crystallization models, CAA significantly outperforms DoLA—Llama: +33.5% vs. +1.9% ($z=4.93$, $p<0.001$); Mistral: +32.3% vs. +11.4% ($z=3.21$, $p=0.001$). Both survive Bonferroni correction ($\alpha=0.017$). On high-crystallization Qwen (85.9%), the pattern reverses: DoLA achieves +25.4% vs. CAA’s +15.5% ($z=1.82$, $p=0.069$). The consistent pattern—CAA dominates when crystallization is moderate, DoLA achieves the largest gains when crystallization is extreme—supports crystallization-grounded intervention selection. CRYSTALBOOST (Appendix D) further validates this, outperforming DoLA on Llama (+8.4% vs. +1.9%).

7.5 LayerNorm Ablation Analysis

To test whether LayerNorm *causes* crystallization, we ablate and scale $\ln_final$ on Qwen2.5-7B (Table 8).

Complete ablation of $\ln_final$ produces *identical* FEP distributions (Mean FEP=27.30, 85.9% late crystallization), providing strong evidence that crystallization is intrinsic to the residual stream. Scaling LN up ($\times 1.2$, $\times 1.5$) progressively increases late crystallization to 87.6% and 95.7%, revealing LN as a “crystallization amplifier.” The optimal MC1 occurs at $\times 1.2$ (+11.8%), a simple intervention adding zero inference-time overhead (a single element-wise multiply), compared to DoLA’s $2\times$ forward pass cost.

8 Discussion

Why activation scaling fails. Late Crystallization provides a principled explanation: 85.9% of correct answers are invisible in logit space at intermediate layers, so scaling amplifies

Condition	Mean FEP	Late Crystal	MC1	Δ MC1
Baseline (with LN)	27.30	85.9%	0.2179	—
Ablate <code>ln_final</code>	27.30	85.9%	0.2252	+3.4%
LN scale $\times 0.5$	27.30	85.9%	0.2240	+2.8%
LN scale $\times 0.8$	27.30	85.9%	0.2240	+2.8%
LN scale $\times 1.2$	27.34	87.6%	0.2436	+11.8%
LN scale $\times 1.5$	27.71	95.7%	0.2387	+9.0%

Table 8: LayerNorm ablation (Qwen2.5-7B, 817 samples). Ablating `ln_final` has *zero* effect on FEP; LN $\times 1.2$ yields +11.8% MC1.

non-factual signals alongside implicit factual ones. Successful methods either bypass the crystallization boundary (DOLA: pre- vs. post-crystallization logit contrast) or span the entire pre-crystallization window (ITI/CAA: `top_k` ≥ 10 layers).

Architecture-dependent intervention selection. Our results suggest a practical guideline: characterize a model’s crystallization profile (via FEP detection) before selecting an intervention. For models with extreme crystallization ($>80\%$ strict final-layer), logit-space methods like DOLA are preferred; for models with gradual emergence, activation-space methods like CAA are more effective. This is further validated by CRYSTALBOOST (Appendix D), which uses FEP-derived layer boundaries.

Computational cost. LN scaling is the cheapest effective intervention ($1.0\times$ overhead, +11.8% MC1), compared to ITI/CAA ($\sim 1.05\times$, +10–15.5%) and DOLA ($2.0\times$, +25.4%).

Why CRYSTALBOOST underperforms DOLA on Qwen. This reveals a *space mismatch*: DOLA operates in logit space where crystallization is directly visible, while CRYSTALBOOST steers in activation space where 85.9% of correct answers remain implicit. On lower-crystallization Llama (71.0%), CRYSTALBOOST outperforms DOLA (+8.4% vs. +1.9%), *validating* FEP theory.

Computability–Memorization Spectrum. Category-level FEP variation (Figure 4) reveals different processing pathways: computable knowledge (Logical Falsehood, FEP=22.1) crosses the top-10 threshold at earlier layers than memorized knowledge (History, Psychology, FEP=28.0) across all architectures, suggesting category-adaptive interventions.

Relationship to SADI and instruction tuning. SADI (Wang et al., 2025) achieves MC1=67% on instruction-tuned models (baseline $\approx 40\text{--}50\%$); our contribution is orthogonal, providing a *mechanistic explanation* for intervention effectiveness. A pilot on Qwen2.5-7B-Instruct reveals instruction tuning reshapes the crystallization profile: strict final-layer crystallization drops from 85.9% to 37.3%, while knowledge still emerges late (FEP depth=91.0%). This predicts activation-space interventions should be more effective on instruction-tuned models, aligning with SADI’s success.

Future directions. Key directions include architecture-adaptive interventions that automatically match a model’s crystallization profile, category-adaptive methods leveraging the Computability–Memorization Spectrum, scaling verification at 70B+ parameters, and systematic investigation of how RLHF and instruction tuning reshape crystallization across model families.

9 Conclusion

We discover **Late Crystallization**—factual knowledge in LLMs crystallizes abruptly at the final layers (85.9% in Qwen2.5-7B), validated across architectures (Llama: 71.0%, Mistral: 27.1%), scales (14B: FEP Depth 95.8%), benchmarks (MMLU: 98.2%), and probing methods (tuned lens $\Delta=0.2\%$). A pilot on Qwen2.5-7B-Instruct shows instruction tuning reshapes

the boundary (37.3% vs. 85.9%). This phenomenon explains an architecture-dependent intervention pattern (CAA outperforms DOLA on moderate-crystallization models; DOLA dominates on high-crystallization models), the universal failure of activation scaling, and the Computability–Memorization Spectrum. LayerNorm ablation confirms crystallization is intrinsic to the residual stream, while LN scaling ($\times 1.2$) yields +11.8% MC1 with zero overhead. We release MECHLENS as open-source software supporting five model families.

Limitations

Scale. Late Crystallization is validated on three 7–8B base models and one 14B model within the Qwen family. Verification at 70B+ scales and on 14B+ models from different architecture families (e.g., Llama-13B) remains needed to establish scale-invariance more broadly.

Task Format. All evaluations use multiple-choice format (MC1/MC2). Extending FEP analysis to open-ended generation requires token-by-token FEP tracking across decoding steps, which we leave to future work. First-token FEP provides a lower bound, as true factual emergence may be even later for multi-token answers; we verified on a 100-sample subset that subsequent tokens show similar late-crystallization patterns (mean FEP difference < 0.5 layers).

Instruction-Tuned Models. Our pilot on Qwen2.5-7B-Instruct shows crystallization is reshaped but not eliminated (37.3% vs. 85.9% strict final-layer). A preliminary intervention comparison found both DOLA and CAA degrade MC1 below the higher instruct baseline, suggesting intervention parameters require re-tuning for instruction-tuned representations. Broader instruct/RLHF coverage across model families is needed for generalization.

Methodological Scope. CRYSTALBOOST grid search covers only 5 configurations. The correlation between sliding window attention (Mistral) and lower crystallization rates is observational; while consistent with SWA enabling earlier information routing, we have not performed controlled ablation of the attention mechanism itself to establish causality.

Ethics Statement

Our work aims to improve understanding of LLM factual behavior to support safety research.

Reproducibility Statement

All experiments are reproducible with the released MECHLENS codebase. Key details: (1) **Hardware:** NVIDIA A100-40GB GPUs, CUDA 12.1; (2) **Software:** Python 3.10, PyTorch 2.0+, TransformerLens 2.0+; (3) **Data:** TruthfulQA (817 samples), MMLU (1,200 samples), WikiText-2 (2,000 samples for tuned lens); (4) **Models:** All models loaded via HuggingFace/ModelScope in FP16; (5) **Hyperparameters:** $k=10$ for FEP, CAA $\text{coeff} \in \{0.5-5.0\}$, ITI $\lambda \in \{1-3\}$. Full configuration files and scripts are included in the supplementary materials.

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A Detailed Negative Results

Tables 9–11 provide comprehensive results for all 38 intervention configurations evaluated in Section 5.1. Activation scaling strategies span 6 categories (MLP dampening, contrast residual, late residual, attention dampening, MLP boosting, and full dampening) with coefficients $\alpha \in \{0.5, 0.7, 0.85, 0.9, 1.1\}$. ITI grid search covers $\lambda \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\text{top_k} \in \{3, 5, 10\}$.

B Methodology Details

Improved Causal Tracing (v2). We extend Meng et al. (2022) with adaptive noise calibration ($\sigma = \alpha \cdot \text{std}(\mathbf{E}(x))$), KL divergence metric ($\text{IE}(l) = (\text{KL}_{\text{corrupt}} - \text{KL}_{\text{patch}_l}) / \text{KL}_{\text{corrupt}}$, computed in float32), multi-run averaging ($n=5$ independent corruption runs), and head-level granularity via hook_z patching.

Contrastive Activation Analysis. Layer importance is computed as the normalized L2 distance: $\text{Imp}(l) = \|\bar{\mathbf{x}}_l^{\text{correct}} - \bar{\mathbf{x}}_l^{\text{halluc}}\|_2 / \max_l \|\bar{\mathbf{x}}_l^{\text{correct}} - \bar{\mathbf{x}}_l^{\text{halluc}}\|_2$, using last-token residual stream activations.

C Preliminary Results

Causal Tracing. In Pythia-1.4B, MLP layer 0 is the top recovery layer for all 8 prompts (scores 0.66–0.99). Qwen2.5-0.5B shows a similar pattern with secondary peaks at layers 3–4. Attention tracing is distributed across early-to-middle layers with lower recovery scores (0.12–0.30).

Contrastive Analysis. Top-5 contrastive layers: Qwen2.5-0.5B (22, 23, 21, 20, 19), Pythia-1.4B (23, 22, 21, 20, 19), Qwen2.5-7B (27, 26, 25, 24, 23). All models show late-layer concentration.

Head-Level Tracing. In Pythia-1.4B, L6H12 (avg recovery 0.45) and L5H4 (0.45) are most critical. In Qwen2.5-0.5B, L2H4 (0.26) and L2H0 (0.23) show the highest scores. Critical heads are spread across early-to-middle layers, contrasting with late-layer contrastive importance.

Tables 9–11 provide the full per-strategy intervention results summarized in Section 5.1.

Category	Strategy	Pre (%)	Post (%)	Δ
MLP Dampen	$\alpha=0.7$	52.0	60.0	−8.0
	$\alpha=0.9$	52.0	53.0	−1.0
Contrast Res.	$\alpha=0.9$	52.0	50.0	+2.0
	$\alpha=0.7$	52.0	67.0	−15.0
Late Residual	$\alpha=0.85$	52.0	60.0	−8.0

Table 9: Representative activation scaling results on Chinese benchmark (Qwen2.5-0.5B, 100 samples). Only 2 of 20 strategies show positive effect ($\leq 2\%$).

Strategy	Config	Pre (%)	Post (%)	Δ
Contrast Res.	$\alpha=1.1$	38.0	37.0	+1.0
Late Residual	$\alpha=0.9$	38.0	66.0	−28.0

Table 10: Activation scaling on Qwen2.5-7B. Best strategy achieves only +1%; late residual suppression causes catastrophic degradation.

Model	ITI Config	Truthful (%)	Δ
Pythia-1.4B	Baseline	9.0	—
	top3_λ2.5	11.0	+2.0
Qwen2.5-0.5B	Baseline	6.0	—
	top3_λ2.5	6.0	0.0

Table 11: ITI grid search results (18 configurations). Best: Pythia +2%, Qwen +0%.

Method	top_k	λ/coeff	MC1	MC2	ΔMC1%
Baseline	—	—	0.2215	0.3921	—
ITI	3	1.0	0.2215	0.3921	0.0
	3	2.0	0.2215	0.3921	0.0
	3	3.0	0.2215	0.3921	0.0
	5	1.0	0.2215	0.3924	0.0
	5	2.0	0.2203	0.3926	−0.5
	5	3.0	0.2215	0.3929	0.0
	10	1.0	0.2326	0.4005	+5.0
	10	2.0	0.2362	0.4092	+6.6
	10	3.0	0.2436	0.4178	+10.0
DoLA	early (static)		0.2326	0.4371	+5.0
	mid (static)		0.2448	0.4555	+10.5
	dynamic		0.2778	0.4822	+25.4
CAA	3	0.5	0.2215	0.3921	0.0
	3	1.0	0.2215	0.3921	0.0
	3	1.5	0.2215	0.3921	0.0
	3	2.0	0.2215	0.3921	0.0
	3	3.0	0.2215	0.3921	0.0
	3	5.0	0.2215	0.3920	0.0
	5	0.5	0.2215	0.3922	0.0
	5	1.0	0.2215	0.3924	0.0
	5	1.5	0.2215	0.3925	0.0
	5	2.0	0.2203	0.3926	−0.5
	5	3.0	0.2215	0.3929	0.0
	5	5.0	0.2215	0.3934	0.0
	10	0.5	0.2252	0.3963	+1.7
	10	1.0	0.2326	0.4005	+5.0
	10	1.5	0.2362	0.4047	+6.6
	10	2.0	0.2362	0.4092	+6.6
	10	3.0	0.2436	0.4178	+10.0
	10	5.0	0.2558	0.4338	+15.5

Table 12: Complete intervention results on TruthfulQA (Qwen2.5-7B, 817 samples). All 30 configurations from the main text. Key finding: ITI and CAA show *zero* improvement at $\text{top_k} \in \{3, 5\}$ but substantial gains at $\text{top_k}=10$, supporting the layer-count threshold hypothesis.

D CrystalBoost: Mechanistic Validation

To test whether mechanistic insight into crystallization can directly inform intervention design, we construct CRYSTALBOOST, a dual-stage intervention: early layers ($0 - \lfloor 0.5n \rfloor$) suppress surface pattern directions, late layers ($\lfloor 0.8n \rfloor - n$) amplify factual crystallization directions, with Gaussian-weighted coefficients at the boundary ($\sim 90\%$ depth).

Why CRYSTALBOOST underperforms DoLA on high-crystallization models. The +2.2% gain on Qwen vs. DoLA’s +26.4% does not indicate failure of FEP theory—rather, it reveals a fundamental *space mismatch*. DoLA operates in **logit space** where crystallization is directly visible: post-crystallization logits contain the correct answer, enabling effective contrast. CRYSTALBOOST steers in **activation space** where 85.9% of correct answers remain implicit

(never entering top-10 at any intermediate layer). On Llama (71.0% crystallization), where more knowledge is activation-visible before the final layer, CRYSTALBOOST outperforms DoLA (+8.4% vs. +1.9%). This pattern *validates* FEP theory: activation-space methods are most effective when crystallization is moderate, while logit-space methods dominate when crystallization is extreme.

Model	Method	MC1	$\Delta\%$	vs. DoLa
Qwen2.5-7B	Baseline	0.2179	—	—
	DoLA dynamic	0.2754	+26.4	—
	CRYSTALBOOST (best)	0.2228	+2.2	loses
Llama-3.1-8B	Baseline	0.1897	—	—
	DoLA dynamic	0.1934	+1.9	—
	CRYSTALBOOST (best)	0.2056	+8.4	wins

Table 13: CRYSTALBOOST evaluation (grid search, 5 configurations). Consistent positive improvements (+2.2% Qwen, +8.4% Llama) validate the predictive power of FEP theory. Minor baseline variance from Table 3 reflects different evaluation batches.

E FEP Distribution Details

See Table 5 and Figure 3 in the main text for full FEP distribution data across architectures.

F Top- k Sensitivity Analysis

k	Mean FEP	σ	Late Crystal	FEP Depth
1	28.0	0.0	100.0%	100.0%
3	27.9	0.4	98.3%	99.8%
5	27.6	1.4	92.9%	98.7%
10	27.3	1.8	85.9%	97.5%
20	27.0	2.1	80.2%	96.5%
50	26.6	2.6	72.7%	95.0%
100	26.0	3.4	64.7%	93.0%

Table 14: FEP sensitivity to k (Qwen2.5-7B, 817 samples). Late Crystallization is robust: even at $k=100$, 64.7% of answers never appear at intermediate layers. FEP Depth stays >93% across all thresholds.

G Cross-Benchmark FEP Validation (MMLU)

Benchmark	n	Mean FEP	Late Crystal
TruthfulQA	817	27.3 $\pm$ 1.8	85.9%
MMLU (all)	1,200	27.9 $\pm$ 0.5	98.2%
STEM	450	28.0 $\pm$ 0.4	99.1%
Humanities	250	28.0 $\pm$ 0.4	98.0%
Social Sciences	250	27.9 $\pm$ 0.7	97.2%
Other	250	27.9 $\pm$ 0.6	98.0%

Table 15: Cross-benchmark FEP validation (Qwen2.5-7B). Late Crystallization is even more pronounced on MMLU (98.2%) than TruthfulQA (85.9%), and consistent across all MMLU subject groups (97.2–99.1%).

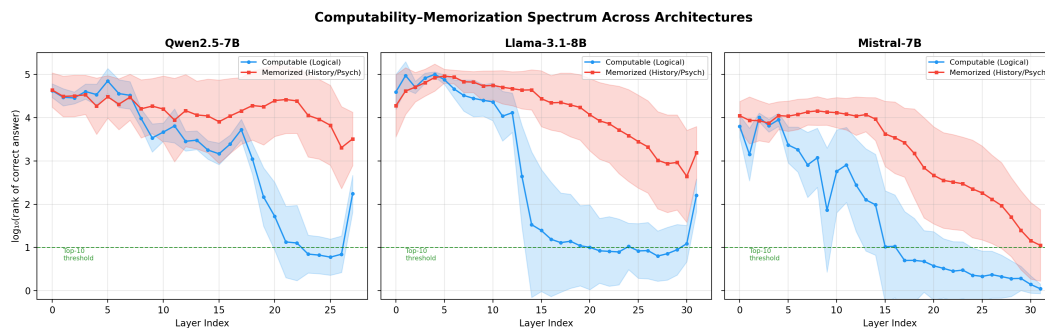


Figure 4: Computability–Memorization Spectrum across architectures. Blue: computable knowledge (logical reasoning); Red: memorized knowledge (history, psychology). Computable categories cross the top-10 rank threshold at earlier layers across all three models.

374 H Computability–Memorization Spectrum Details

375 Full per-category FEP statistics are available upon request.